

THE LITTLE HOUSE THAT COULD

A new Habitat for Humanity-built home is a net producer of energy, having met and exceeded its zero-energy goal.

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In October 2005, Amy Whalen and her two young boys moved into their new Habitat for Humanity home near Denver, Colorado. In doing so, they became partners with DOE in a case study aimed at understanding how to create affordable zero-energy homes in cold climates. The home was a result of a collaboration between the National Renewable Energy Laboratory (NREL), our employer, and Habitat for Humanity of Metro Denver.

A zero-energy home (ZEH) is designed to produce as much energy as it consumes over the course of a full year. The home uses the utility power grid for storage—delivering energy to the grid when the PV system is producing more energy than is being used in the home and drawing from the grid when the PV system is producing less energy than is needed in the home. This approach eliminates the need for battery storage in the home, thereby reducing the cost, complexity, and maintenance of the solar-electric system.

Reaching zero energy within the affordable housing sector in cold climates can be a challenge. With the zero-energy goal in mind, we designed a 1,200 ft², three-bedroom Denver zero-energy home that carefully combines envelope efficiency, efficient equipment, appliances and lighting, and passive- and active-solar features to reach the zero-energy goal. The home was designed using an early version of the BEopt building optimization software, a computer program designed to find optimal



The NREL/Habitat Zero-Energy Home combines performance, ease of construction, and low cost while maintaining the zero-energy goal. Notice the double-stud walls that leave more room for insulation than single-stud walls do.

building designs along the path to zero net energy. Additional analysis was provided by DOE-2, a widely used and accepted freeware building energy analysis program that can predict the energy use and cost for all types of buildings. This engineering approach was tempered by regular discussions with Habitat construction staff and volunteers. These discussions weighed the applicability of the optimized solutions to the special needs and economics of a Habitat house—moving the design toward simple, easily maintained mechanical systems and volunteer-friendly construction techniques.

Cold Climate Challenge

Homes account for 37% of all U.S. electricity consumption and 22% of all U.S. primary energy consumption. This makes home energy reduction an important part of any plan to reduce U.S. contribution to global climate change. The goal of DOE's Building America program is to create commercially viable zero-energy homes by 2020. The NREL/Habitat ZEH project presents a case study in reaching that goal within the affordable housing sector in cold climates. Zero energy is especially important in this sector, where increasing energy cost can take a high toll on homeowners with limited economic resources. A zero-energy home

guarantees long-term energy cost stability for the homeowner.

From its inception, the NREL/Habitat ZEH project focused on finding a balance between engineering ideals and real-world practicality. The team that designed the home included two NREL building energy researchers; two Habitat staff members (the construction manager and the real estate development manager); and two Habitat energy subcommittee volunteers. The NREL engineers

1. It should use zero net energy.

Zero net energy can be defined in terms of site energy (used at the building site) or source energy (sometimes called primary energy). For electricity purchased from a utility, the source energy used to produce and distribute the electricity is typically about 3 times as much as the delivered electricity. From a societal point of view, source energy better reflects the overall consequences of energy use. The home was designed to

3. It should take advantage of Habitat volunteer labor.

When considering construction alternatives, we took into account that Habitat's approach to building with volunteer labor presents a unique opportunity to reduce building cost. Construction techniques that were volunteer-friendly and tended toward low material cost were favored.

4. Tradeoffs for zero energy should be done at full material cost.

Although some of the equipment in the house was donated or bought with grants at no cost to Habitat, we considered the full value of these items to find the balance between efficiency and PV production.

5. No special operation of the building should be needed.

After construction, this house was sold to a Habitat family. It was important to the design team to have the home's energy-efficient attributes as invisible to the family as possible. From the family's perspective it should be a normal home; the owner should not have to do anything extra to operate the building.

6. No prototypes!

We designed the home with off-the-shelf proven technologies available in the marketplace today. Although optimal research systems were discussed as part of the design process, the final design aimed to use commercially available products to come as close as possible to the ideal. Because the home is expected to outlive all of its mechanical systems, we wanted these systems to be easily replaceable by technicians the owners could find in the local yellow pages.

7. Keep it simple.

Many of the recently designed zero-energy homes include complicated interconnected mechanical systems designed to maximize renewable energy use and distribution. We too were often tempted in this direction. We tempered this temptation by continually striving to keep it simple. We believe a simpler



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The 1,200 ft², three-bedroom home carefully combines envelope efficiency, efficient equipment, appliances, and lighting, and passive and active solar features to reach the zero-energy goal. The south-facing windows are well shaded by the roof overhang.

made suggestions based on modeling results and analysis and presented them to the design team, who then grounded the discussion with practical concerns and insights. This mix of perspectives led to a design that balances energy performance, ease of construction, and low cost while maintaining the zero-energy goal.

Project Design Criteria

A Habitat for Humanity house presents an unusual opportunity. Thanks to volunteers, much of the labor comes at no cost, and it is possible to get some of the equipment donated or at reduced cost. We established the following criteria for the home design:

meet the definition of zero energy established by DOE's Building America residential energy efficiency research program. It must have predicted zero net source energy consumption over the course of a year, using typical meteorological year (TMY) weather data, and Building America (BA) Benchmark assumptions on occupant behavior based on average U.S. behavior in terms of temperature set points, miscellaneous electric loads, and hot water use.

2. It should be replicable by Habitat for Humanity.

Construction techniques and energy efficiency technologies were vetted for the probability that they could be repeated in future homes.

system will have fewer problems and a greater chance of longevity.

Final Home Design

The home design process used a combination of computer simulations and heuristic judgment. Three simulation tools were used both sequentially and iteratively during the design process. These tools were TRNSYS Transient System Simulation software, DOE-2 Building Energy Model, and BEopt Building Energy Optimization program.

The envelope design began by looking at Habitat Metro Denver's standard home plans (see "Habitat ZEH Attributes"). We sorted

Table 1. ZEH Total Energy Use February–July 2006

	Site kWh	Site Therms	Source MBtu
Total natural gas use		23.6	2.40
Total electricity use	1,701		18.34
Total PV production	3,317		35.77
Net electricity used	-1,616		-17.42
Net source energy			-15.02

Table 2. Hot Water Energy Use for April 2006

	Site kWh	Site Therms	Source MBtu	Percentage
Natural gas for water heating		0.3	0.03	
Solar contribution to water heating		3.4	0.35	
Solar water pump	9		0.09	
Solar savings fraction (w/o pump energy)				92%
Site solar savings fraction (w/pump energy)				84%
Source solar savings fraction (w/pump energy)				67%

these plans for their applicability to the site and adaptability to passive-solar design. A standard three-bedroom 26 ft x 46 ft design with a crawlspace was chosen. The floor plan was mirrored from its original design to accommodate the site.

Motivated by the BEopt simulation recommendation for a superinsulated envelope, the design team considered a wide variety of approaches, including structural insulated panels (SIPs), insulated concrete forms (ICFs), straw bale, and double-stud walls. We chose a double-stud wall with fiberglass batt construction because it has low material

Habitat ZEH Attributes

The standard Habitat Metro Denver home includes features such as 2-inch blueboard insulation on the interior face of crawlspace walls, 6-mil plastic sheeting on crawlspace floors, and 2-foot eaves overhangs. The zero-energy house has those features and more, including:

Superinsulated building envelope

- Double stud wall construction, 1ft overall wall thickness
- Fiberglass insulation: R-40 walls, R-60 ceiling, R-30 floor
- Air tight with continuous air and vapor barrier
- Air tight outlets in exterior walls

Passive-solar design

- Expanded southern window area with summer shading overhangs
- Reduced east and west window area

- Low-e windows throughout (U-0.23–U-0.3)
- Orientation specific glazing: high SHGC (0.58) on the south; low SHGC (0.27) on west, north, and east

Solar water heating with efficient back-up heater

- 96 ft² of collectors on roof with 200 gal bladder tank in the utility room
- Back-up water heating from high efficiency tankless water heater
- 4 kW photovoltaic system

- Grid-tied with net metering, no batteries
- Sized to meet all electrical loads annually and offset natural gas use

Fresh air ventilation with energy recovery

- Energy recovery ventilation (ERV) system with ducting inside conditioned space
- ERV is equipped with high efficiency ECM motors

Simple space-heating system

- Thermostatically controlled ductless direct

- vent natural gas space heater in the living/dining room
- Small thermostatically controlled electric baseboard heaters in each bedroom

Other features

- All mechanical equipment is within the conditioned space
- Light-colored roofing shingles and increased attic ventilation reduces attic overheating
- Energy Star appliances
- 100% CFLs

costs, uses familiar volunteer-friendly construction techniques, and uses proven construction techniques. Raised heel trusses were designed to accommodate 2 feet of blown-in fiberglass in the attic, giving the top of the thermal envelope an R-60 rating. The floors are insulated to a nominal R-30.

The walls consist of an outer 2 x 4 structural stud wall 16-inches OC with R-3 fiberglass batts in the cavities. Spaced 3-½ inches inside this wall, a second 2 x 4 stud wall 24 inches OC was built. Additional R-13 fiberglass batts were placed horizontally in the space between the stud walls and vertically in the interior wall cavities. An outer vapor-permeable house wrap and fiber cement siding and an inner poly vapor barrier and dry-wall complete the nominal R-40 assembly. The actual whole-wall R-value of this wall will be much closer to its nominal value than would be the case with a single-stud wall, because the thermal shorting of the studs is broken by the insulation in the space between the double-stud walls.

Double-glazed, low-e high-SHGC glass was chosen for the southern windows. The U-value of this glass was 0.3, and the SHGC was 0.58. Double-glazed clear windows would have provided more solar heating, but also would have increased the overheating potential and would have lacked the thermal comfort and ultraviolet screening benefits provided by the glass that we used. Double-glazed low-e low-SHGC glass was used for the east, west, and north windows. The U-value of this glass was 0.22 and the SHGC was 0.27.

Ventilating and Heating the Home

Because we intended to build the home with very low air leakage, a mechanical ventilation system was required. To provide fresh air to the home while minimizing energy losses, we chose to use a balanced energy recovery ventilation (ERV) system. The

ERV exhausts air from the kitchen and bathroom and supplies fresh air to the living room and the bedrooms. The warmth of the exhaust air is used to heat the incoming fresh air. This significantly reduces the heat loss due to ventilation. We chose an ERV with efficient electronically commutated motors.

Having a very low design heating load is a blessing and a challenge. The blessing is obvious—it takes very little energy to keep this home warm! The challenge is that most commonly available heating

comes primarily from fossil fuels. So the home is consuming fossil fuels when it is using electricity and offsetting that consumption when it is producing excess PV electricity (see Table 1 and Figure 1). Much the same holds true for a ZEH that consumes natural gas. The PV system is sized to produce an excess of electricity to offset the natural gas used. The goal is net zero source energy use.

The economics of the use of natural gas or all-electric varies, and there are advantages and disadvantages with both. For an all-electric home using resistance heating, the all-electric approach requires a larger PV system and is substantially more expensive. The all-electric approach has the advantage of eliminating the monthly fixed cost of a natural gas hookup, which is about \$9 per month in the Denver area. However, the Habitat ZEH design team decided to use natural gas in the home, reducing the size of the required PV array by 1.1 kW, and to take a hybrid approach to space heating.

The space-heating system combines a point source direct-vent natural gas furnace in the living and dining area of the home with small baseboard electric-resistance heaters in the three bedrooms. This approach is relatively inexpensive and elegantly simple, and it provides zone heating, since each appliance has its own independent thermostat.

Although we ruled out a solar combisystem (providing both space heating and water heating) as being too complex, the results of the early BEopt runs convinced us to incorporate a high solar-savings fraction solar water-heating system into the home design (see Table 2). We used TRNSYS to do parametric studies to design the system. We found that mounting the collectors at the roof pitch rather than raising them to their optimal angle incurred only a small annual energy penalty. We found that a 96 ft² collector area with 200 gallons of water storage

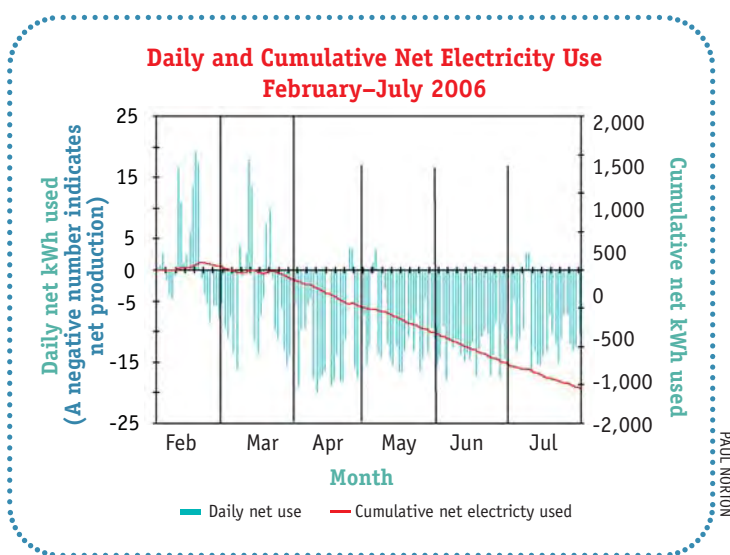


Figure 1. During the summer the home has produced substantially more energy than it consumed. This excess production will be more than enough to offset the winter energy consumption.

systems are oversized for this home and the low heating energy needs cannot justify a complicated or expensive system. We considered a wide variety of high-efficiency heating systems for the home. These included active-solar thermal with a radiant floor, baseboard heaters, or a fan air coil in the ERV supply; a ground-coupled heat pump (GCHP); a point source natural gas furnace (with no duct system); or a hydronic system to distribute the heating. The compact size and superinsulated shell of the Habitat ZEH reduced heating needs to so low a level that the cost of the GCHP was not justified.

The use of natural gas for heating, cooking, and clothes drying in a ZEH is somewhat controversial. There are those who believe that since a ZEH exports only electricity, it must consume only electricity. However, in most of the United States, the electricity consumed

would result in an 88% annual solar-savings fraction using TMY2 weather data and BA Benchmark hot water use. This solar-savings fraction includes pump energy use.

We specified a natural gas tankless water heater as a backup to the solar system. Unlike tank water heaters, the tankless system uses no heating energy when the solar water tank is at or above the 115°F hot water delivery temperature, although it may use electricity in standby. The disadvantage of using the tankless system is the added cost compared to a tank system. We considered using the tankless water heater for space heating also, but ultimately decided to use separate systems to avoid the complexity of the combined system.

Sizing the PV System

Once all possible energy loads in the house were significantly reduced, the PV system was sized to meet the remaining electricity needs and offset the expected natural gas use. In a similar home built to BA Benchmark standards, about one-quarter of the energy in the home is consumed by lighting, appliances, and miscellaneous electric loads (LAME loads). We reduced the lighting load by using CFLs throughout the home and reduced the appliance load by using Energy Star appliances (see Figure 2). This leaves the miscellaneous electric loads (MELs), which include everything that might be plugged in by the occupants: TV, hair dryer, toaster oven, computer, aquarium, and so on. Because all other loads have been dramatically reduced, the MELs in the Habitat ZEH are expected to consume 57% of all energy used annually. Although the Building America program is pursuing research into ways to reduce these loads, they are currently out

of the control of the home designer. Furthermore, these loads are highly unpredictable and vary substantially from household to household. So the ZEH designer is faced with sizing a PV system for a home where the largest load is really not known with any accuracy.

The BA Benchmark includes assumptions that we used to estimate the MELs and size the 4kW PV system. These assumptions are based on the best

From the beginning of February to the end of July 2006, the PV system produced 1,600 kWh more than the electricity used in the home. Accounting for the 30 therms of natural gas used for space heating, backup water heating, and clothes drying, the home produced about 72% more source energy than it consumed during this period. It is clear that this home not only will reach its goal of zero energy use, but will be an annual net energy producer. This is primarily because the occupants use less energy than predicted by the BA Benchmark assumptions that were used to size the PV system.

Although the home is a net energy producer, the energy bills are not zero. The owners must pay for the natural gas used, and there are fixed charges for the electric and natural gas connections each month. From October 2005 to May 2006 the energy bills for the house averaged \$18.25 per month. The fixed charges averaged 80% of the total energy bill.

The solar water-heating system has successfully delivered high solar-savings fractions on a thermal basis. For example, the solar-savings fraction in

April 2006 was 92%, not counting the energy consumed by the pump. When the electricity used by the pump is taken into account, the solar-savings fraction drops to 84% on a site energy basis and 67% on a source energy basis.

From February to July 2006, the miscellaneous electric loads were over 60% of the total energy use in the home. The MELs in the Habitat ZEH are actually lower than the BA Benchmark, but the other energy use in the home is so small that the MELs are a large percentage of the total.



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**Cumulative Electricity End Uses
February–July 2006**

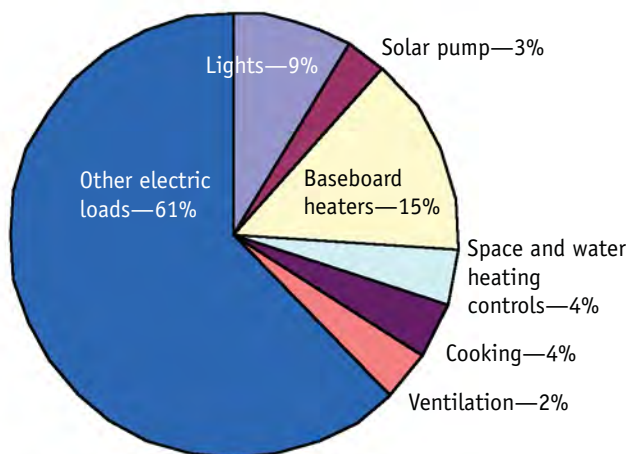


Figure 2. The cumulative electricity use is dominated by the "other electric loads" which include refrigerator, entertainment center, microwave, coffee maker, computer, clocks, and so on, that are outside of the control of the building designer.

available nationwide studies of energy use. So the home's PV system is sized with the assumption that it will be occupied by a "typical" American household. If the actual household and weather are typical, the home will achieve zero energy. If the household or weather is atypical, the home may not achieve zero energy, or it may be a net producer.

Home Performance

We installed a data acquisition system in the home in January 2006 and will monitor the home's performance for at least one year. Blower door testing by Paul Kreischer of Lightly Treading, Incorporated, yielded a natural infiltration result of 0.15 ACH, which is quite tight. The home was Energy Star rated at 95.